

Multi-Stage Solicitation

Chained elicitation markets, and lessons from the microprediction platform

Peter Cotton

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Abstract

Deployed forecasting contests almost always run one level of competition against one internal model. The microprediction platform was built the other way: it chained elicitation mechanisms, one market’s probabilistic output becoming the message or the settlement of the next. Community percentiles from univariate z-streams were themselves the object of further games; bivariate and trivariate dependence streams priced copulas; a stacked-lottery design composed monotone calibration maps contributed by competing algorithms. This note reports that design and the theory it rests on. The single-stage dictionary — proper scores, market makers, and pools as one convex object — is a companion paper ([Cotton 2026a](#)); here the object is the chain. The message type deployed contests collect is a finite cloud of samples, and the sample-native forms of the composition operators are given; multi-variate settlement factors into marginal stages and a rank-settled copula stage; and a residual market behind a conformal predictor collects the predictor’s discarded conditional information as bankroll growth. The platform is retired; its nearest live successor, monteprediction, runs one stage of the chain repeatedly.

1 Introduction

Deployed forecasting contests are thinner than the theory allows: with few exceptions they run one level of competition against one internal model. Numerai pools staked submissions into a single stake-weighted meta-model and pays each forecast its marginal contribution to it ([Craib et al. 2017](#)); CrunchDAO blends each contest into one ensemble; the IARPA prediction polls aggregated forecasts by track record, comparing favourably with head-to-head markets ([Atanasov et al. 2017](#)), with reputation rather than wealth as the threaded state. In every case, one pool and one internal aggregate.

The microprediction platform was built the other way. A market’s *probabilistic* output — a distribution, a percentile, or a rank emitted by one stage — became the message or the settlement transform of the next, so calibration and dependence were themselves the subject of further games. Univariate streams spawned z-streams of community percentiles; bivariate and trivariate dependence streams priced copulas; and a stacked-lottery design composed monotone calibration maps contributed by competing algorithms, presented at MIT CSAIL in 2020 ([Cotton 2020](#), slides 29-31; [Cotton 2022](#)). The platform is retired. Its nearest live successor, monteprediction, is a weekly self-funding pool over million-scenario joint submissions in eleven dimensions with wealth threaded across rounds since January 2024 ([Cotton 2024b](#)): one stage of the chain, repeated, rather than a chain.

This note is a report on that design. The mechanisms compose because each is a stateful transducer over one message type, a distributional belief, with wealth as the threaded state; the single-stage dictionary that generates them — proper scores, market makers, and pools as one convex object — is a companion paper (Cotton 2026a), and §2 recalls only what the chain needs. The rest is the multi-stage account the platform demanded: the finite sample cloud as the message type deployed contests actually collect (§3), the factoring of a multivariate settlement into marginal stages and a rank-settled copula stage (§4), and the residual chain behind a conformal predictor whose degenerate case leaves an extractable rent (§5). Play is stagewise throughout: no participant deviates across stages; the cross-stage game is largely open (§6).

2 Stages and their composition

A *stage* is a transducer (Mealy machine) over one message type. Let Dist be the set of distributional beliefs; a stage carries a wealth state w , consumes participant reports in Dist^m together with, where it settles, a realized outcome x , and emits an aggregate belief and a transfer vector,

$$M : (\text{Dist}^m, w, x) \mapsto (\text{Dist}, w', \pi).$$

A scoring rule is the transfer component of a stage; a market maker is a stage whose state is the inventory vector and whose emitted belief is the price; an opinion pool is a stage with no outcome argument and zero transfers; a parimutuel is a stage whose transfers are a pot split. Composition wires the belief output of one stage to the belief inputs of the next while state and transfers thread through.

The single-stage operators — Sequentialise (a proper score run against a wealth state is a cost-function market maker), Pool, Ensemble, Merge, and Conjugate — and the convex dictionary behind them are the companion algebra (Cotton 2026a). Two operators do the chaining, and their guarantees are what let a pipeline be trusted.

Residual. Let a stage emit the aggregate F_1 for an outcome Y , and let a second market elicit a distribution for the residual $U = F_1(Y)$, settling at the realised $u = F_1(y)$. If F_1 is the true conditional law then U is uniform (the probability integral transform (Cotton 2026a, Prop. 4)) and the second market has nothing to price; whatever structure remains in the residual is the second stage's edge. The corrected forecast composes the two reports.

Proposition 1 (the correction is a multiplicative reweighting). *Let F_1 be strictly increasing onto $(0, 1)$ with density $p_1 > 0$ and let the residual market's consensus be a distribution H on $[0, 1]$ with $H(0) = 0$, $H(1) = 1$ and density g . The composed forecast $F = H \circ F_1$ has density*

$$p(y) = p_1(y) g(F_1(y)),$$

so $\log p(y) = \log p_1(y) + \log g(u)$ with $u = F_1(y)$: the chain's log score is the sum of stage log scores, and the residual stage is paid by a proper score on u alone.

Proof. Chain rule: $F'(y) = g(F_1(y)) p_1(y)$; take logarithms. The residual score $\log g(u)$ is the logarithmic score (Cotton 2026a, Thm 1) applied to the report g and outcome u , hence strictly proper for the law of U . ■

Multiplying the density by a ratio fitted to what the current model gets wrong is the functional-gradient step of boosting under log loss (Mason et al. 1999; Friedman 2001), so a chain of residual markets is stagewise boosting with wealth as the learning rate. What

Proposition 1 does not settle is the game across stages: who funds the residual pot, and whether a forecaster free to enter both stages prefers to withhold information from the first and sell it to the second (§6).

Spec. Serialise a pipeline to data and search over it; the mechanism analogue is a market over pipelines. Also open.

Stagewise play. Call a profile of reports a *stagewise equilibrium* if no participant gains by a deviation confined to a single stage, all other stages' reports held fixed *and the inputs and settlement transforms of every other stage clamped at their pre-deviation values*. This is the pipeline version of the myopic-trader assumption standard in the market-scoring-rule literature (Hanson 2007; Chen and Vaughan 2010). The clamp is not cosmetic: a stage's output is wired into later settlement transforms (the residual point $u = F_1(y)$, or a rank vector in a copula stage), so an upstream deviation moves downstream payoffs even when every downstream report is held fixed.

Proposition 2 (single-stage guarantees compose under clamped stagewise play). *Suppose each stage of a pipeline, taken in isolation with its inputs and settlement transform fixed, makes the truthful report a best response: strict propriety for externally funded scoring stages (Cotton 2026a, Thm 1), the sequential scoring (Cotton 2026a, Thm 2) for market stages, the price-taking pot-split analysis of the point-cloud paper (Cotton 2026c) for parimutuel stages, and the residual score of Proposition 1 for correction stages. Then truthful reporting at every stage is a stagewise equilibrium of the pipeline. If moreover no participant reports to a stage upstream of one in which they hold a position (disjoint stage membership suffices), the clamp is vacuous for every feasible deviation, and truthful reporting survives unrestricted single-stage deviations.*

Proof. With the other stages' inputs and transforms clamped, a deviation confined to stage k changes the deviator's payoff only through stage k 's transfer map, and the stage- k hypothesis makes the truthful report a best response. For the second claim: a stage- k deviation moves stage k 's transfer and, through stage k 's output, transfers strictly downstream of k ; a deviator with no downstream position collects none of the latter, so the propagation is payoff-irrelevant to them. ■

Proposition 2 is not a Nash equilibrium of the composed game. When the same participant reports upstream and holds exposure downstream, the deviation propagates through the settlement transform, and a downstream stake is a derivative written on an upstream settlement, with the attendant incentives to distort the underlying (Kumar and Seppi 1992; Jarrow 1994; Hanson and Oprea 2009; Ostrovsky 2012). Disjoint membership, zero downstream exposure for upstream reporters, or exogenous freezing of the settlement transforms restore the proposition; the dynamic game without them is outside this note's scope.

The rest of this note reports the two message types the platform chained on — sample clouds (§3) and copula ranks (§4) — and the residual chain that a conformal predictor degenerates (§5).

3 Samples as messages

The message type has so far been a distribution given exactly. Deployed contests collect something rougher: a finite cloud of samples, smoothed by the contest into a density before settlement. The companion point-cloud paper gives the sample-based elicitation result the algebra needs (Cotton 2026c):

Proposition 3 (sample-based elicitation; Cotton (2026c), Thms 1-2). *Score the bandwidth- h kernel density estimate of a submitted cloud by the logarithmic score. Settled at the raw outcome,*

the optimal cloud is drawn from a deconvolution of the belief when one exists (for Gaussian beliefs and kernels with belief variance exceeding h^2 , the belief with h^2 removed from the variance). Settled at the outcome jittered by the same kernel, truthful sampling is optimal, and strictly so whenever the kernel's characteristic function is nonvanishing on a dense set: the smoothing channel is injective on laws.

The proof is in the companion paper. With jittered settlement the cloud is a valid message, and the operators of the algebra ([Cotton 2026a](#)) act on clouds directly.

- *Conjugate* acts pointwise: the pushforward of a cloud under ψ is ψ applied to each sample, with no Jacobian computed anywhere. This is the practical reason to prefer sample messages: transformation of densities requires calculus, transformation of clouds requires arithmetic.
- *Pool* (linear) is a wealth-weighted union: sample from participant i 's cloud with probability proportional to w_i . The linear pool is the sample-native aggregate; the logarithmic pool has no comparably clean sample form, since multiplying densities requires importance weights.
- *Residual* acts by ranking: pass each sample through the upstream CDF. Composition of monotone maps, the stacked-lottery operation, is again pointwise on samples.
- *Sequentialise* is the nearest-the-pin pool itself: the market's state is the field's aggregate cloud.

One caveat governs the order of operations. Smoothing and conjugation commute only for affine maps: for scalar $\psi(x) = ax + b$ and a Gaussian kernel, the KDE of the mapped cloud at bandwidth $|a|h$ equals the pushforward of the bandwidth- h KDE (in higher dimension the bandwidth matrix must transform with the map), and for nonlinear ψ no bandwidth makes this an identity. The smoothing seam and its jitter must therefore be applied in the settlement coordinates, after all transforms, which in pipeline terms says the KDE stage belongs immediately before settlement and nowhere else.

4 Margins and copulas

A multivariate settlement can be factored. By Sklar's theorem ([Sklar 1959](#)) a joint law with continuous margins $F_1, \dots, F_d$ and copula C has density

$$p(x) = \prod_{i=1}^d f_i(x_i) \cdot c(F_1(x_1), \dots, F_d(x_d)),$$

with c the copula density on $[0, 1]^d$.

Proposition 4 (the log score factors). *For a joint report assembled from margin reports f_i and a rank-stage report c , a density on $[0, 1]^d$,*

$$\log p(x) = \sum_{i=1}^d \log f_i(x_i) + \log c(u), \quad u_i = F_i(x_i),$$

so a pipeline of d margin stages and one rank stage settled on the rank vector u pays every stage a logarithmic score of its own object, and the chain's log score is the sum. Given the margin reports, the rank stage's score is strictly proper for the law of the rank vector. That law has uniform margins, and is

the copula of the joint, iff the margin reports are correct; keeping the rank stage's report class as arbitrary densities on $[0, 1]^d$ keeps the stage closed under wrong upstream reports.

Proof. Take logarithms in Sklar's density factorization; the summands are exactly the stage scores. Strict propriety of the rank stage is the logarithmic score's strict propriety (Cotton 2026a, Thm 1) applied to densities on $[0, 1]^d$, the transform $x \mapsto u$ being fixed once the margin reports are given. The final clause is Sklar's theorem applied to the true joint law. ■

This is the two-stage estimation logic of inference functions for margins (Joe 1997), and the factorization underlies out-of-sample copula comparison (Diks et al. 2010); here it is read as a market design. The rank stage is the multivariate Residual operator of §2: under correct margins the rank vector has uniform margins and its joint law is the dependence structure alone, so a rank-settled market elicits dependence separately from level. When the margins are correct the settled object is invariant to monotone transformations of the coordinates; the settlement transform itself moves with the reported margin maps. The microprediction platform ran exactly this factoring: alongside univariate z-streams it operated bivariate and trivariate streams in which community-implied percentiles were embedded by a space-filling curve into one settled scalar (Cotton 2022), a copula market in production. Sample messages compose with the factoring: rank each coordinate of a submitted cloud through the margin CDFs and the result is a cloud on $[0, 1]^d$ for the copula stage, with the §3 seam discipline applied at that stage's settlement.

5 A residual chain: betting against a conformal predictor

The purest deployed chain is a residual market run behind a point predictor. A point predictor leaves a residual R ; re-leveling its marginal law is conformalization, and a market on the residual rank prices whatever conditional structure the re-leveling ignores. A single-shape conformal predictor, one that applies the same residual law at every input, prices the residual pool flat: its marginal coverage is exact and its conditional information is discarded. An entrant who conditions on the input X collects that discarded information as bankroll growth, at the rate $I(R; X)$, the mutual information between residual and input, while the band's marginal coverage stays exact. Marginal coverage is the break-even statement; the conditional information is the rent.

The full account — the pool payoff, the Kelly-Breiman growth identity, the Gaussian rate $-\frac{1}{2} \log(1 - \rho^2)$, and the anytime-valid measurement that turns the rent into a test — is a standalone note (Cotton 2026b). The mechanism ran in production: the microprediction nearest-the-pin pool (Cotton 2022) paid each entry its sample density at the realised value relative to the field's, and the MidOne contest (CrunchDAO 2024) priced an explicit residual density (Cotton 2024a) the same way. A conformal predictor entering such a pool is the participant that prices flat in X ; its better-informed competitor's bankroll is the rent.

6 Open problems

1. *Conservation of edge.* Wealth conservation is automatic, a sum of stage-level zeros, and is not the invariant. The invariant that can fail is the edge a truthful participant holds over the price, $D_G(q, \pi)$. For the logarithmic score the edge is $\text{KL}(q \parallel \pi)$ and the data-processing inequality settles it: an interface channel cannot increase it, and preserves it exactly when the channel is sufficient for the pair (Csiszár 1967). No such inequality holds for a general

Bregman edge. For the quadratic generator, merging the first two of three outcomes sends $\pi = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ and $q = (\frac{13}{30}, \frac{13}{30}, \frac{2}{15})$, with $D_G(q, \pi) = \frac{3}{50}$, to a pair with $D_G(qK, \pi K) = \frac{2}{25}$: coarsening increased the Brier edge. Open: characterize the generators and channels for which $D_G(qK, \pi K) \leq D_G(q, \pi)$, and quantify the contraction or amplification of edge through a given interface.

2. *Residual markets.* The single-step correction is a multiplicative reweighting of the upstream density (Proposition 1); a chain of residual markets is stagewise boosting with wealth as the learning rate. Open: the microstructure (who funds each residual pot, and when it settles relative to the stage before), whether a forecaster free to enter several stages prefers withholding information upstream to sell it downstream, and whether the iterated correction converges to the true conditional law as boosting does. Split-conformal prediction is the one-participant degenerate case, with its rent $I(R; X)$ as the value left on the table (Cotton 2026b). The closest live analogue is in point prices: a virtual bid in a two-settlement electricity market is written on the day-ahead stage’s pricing error, and its convergence bidders are the residual stage’s informed entrants (Jha and Wolak 2023). The distributional version is the gap.
3. *Copula markets.* A rank-settled stage elicits dependence separately from margins (Proposition 4), and under correct margins its settled object is invariant to monotone transformations of the coordinates. The racetrack’s win and exotic pools price margins and joint orders in parallel books on a finite outcome space, with consistency left to arbitrage and the Harville map as the bridge (Harville 1973; Hausch et al. 1981); the rank-settled stage differs by construction, chaining through the reported margins so the dependence market cannot disagree with them. Open: the equilibrium when the same participants trade margin and copula stages, and the choice of embedding for $d \geq 2$, the space-filling curves as deployed against the random one-dimensional projections of the point-cloud paper (Cotton 2026c).

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